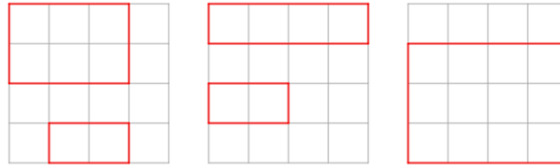


Question 14 – Stage 1



A 4 by 4 square grid contains 16 unit squares and, with gridlines for edges, a total of 35 landscape rectangles (i.e. a rectangle whose width is greater than its height).

Five of these 35 rectangles are shown above.

A 45 by 45 square grid contains 2025 unit squares.

With gridlines for edges, how many landscape rectangles does it contain?

Working

```
In [1]: totalPossibleRectangles = 0
        gridSize = 45

        # For rectangle of width x and height y
        for x in range(2, gridSize+1):
            for y in range(1, x):
                # Number of x positions * number of y positions
                totalPossibleRectangles += (gridSize - x + 1) * (gridSize - y + 1)

        print(f"Total Number of Rectangles: {totalPossibleRectangles}")

Total Number of Rectangles: 519915
```